

4.6.3 Magnetism and electromagnetism

This topic starts with a study of the magnetic fields around permanent magnets and the Earth. The study of the Earth's magnetism can be used to provide clues to the planet's internal structure. Electric currents produce magnetic fields. Forces produced in magnetic fields can be used to make things move. This is called the motor effect and is how an electric motor creates movement.

4.6.3.1 Magnets

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the attraction and repulsion between unlike and like poles for permanent magnets and describe the difference between permanent and induced magnets.	The poles of a magnet are the places where the magnetic forces are strongest. When two magnets are brought close together they exert a force on each other. Attraction and repulsion between two magnetic poles are examples of non-contact force. A permanent magnet produces its own magnetic field. An induced magnet is a material that becomes a magnet when it is placed in a magnetic field. Induced magnetism always causes a force of attraction. When removed from the magnetic field an induced magnet loses most/all of its magnetism quickly.	

4.6.3.2 Magnetic fields

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the characteristics of the magnetic field of a magnet, showing how strength and direction change from one point to another.	The region around a magnet where a force acts on another magnet or on a magnetic material (iron, steel, cobalt and nickel) is called the magnetic field. The force between a magnet and a magnetic material is always one of attraction. The strength of the magnetic field depends on the distance from the magnet. The field is strongest at the poles of the magnet. The direction of the magnetic field at any point is given by the direction of the force that would act on another north pole placed at that point. The direction of a magnetic field line is from the north (seeking) pole of a magnet to the south (seeking) pole of the magnet.	WS 2.2 Draw the magnetic field pattern of a bar magnet and describe how to plot the magnetic field pattern using a compass.

4.6.3.3 The Earth's magnetism

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how the behaviour of a magnetic compass is related to evidence that the core of the Earth must be magnetic.	A magnetic compass contains a small bar magnet. The Earth has a magnetic field. The compass needle points in the direction of the Earth's magnetic field. The Earth's magnetic field is probably caused by movements in the liquid, iron-rich part of the outer core of the Earth. The slow changes to the positions of the magnetic north and south poles, and the way that the field reverses its direction from time to time, show that the magnetism of the core is dynamic and not static. The intervals between reversals are not uniform. The last reversal happened about 800,000 years ago.	WS 1.3 Explain why the data needed to answer a scientific question, in a given context, may not be available because of matters of scale and complexity.

4.6.3.4 The magnetic effect of an electric current

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe how to show that a current can create a magnetic effect and describe the directions of the magnetic field around a conducting wire.	When a current flows through a conducting wire a magnetic field is produced around the wire. The shape of the magnetic field can be seen as a series of concentric circles in a plane perpendicular to the wire. The direction of these field lines depends on the direction of the current.	WS 1.2, 3.1 Draw the magnetic field pattern for a straight wire carrying a current (showing the direction of the field). WS 1.2 Use the 'right-hand grip rule' to predict the direction of the field.

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that the strength of the field depends on the current and the distance from the conductor, and explain how solenoid arrangements can enhance the magnetic effect.	The strength of the magnetic field depends on the current through the wire and the distance from the wire. Shaping a wire to form a solenoid increases the strength of the magnetic field created by a current through the wire. The magnetic field inside a solenoid is strong and uniform. The magnetic field around a solenoid has a similar shape to that of a bar magnet. Adding an iron core increases the magnetic field strength of a solenoid. An electromagnet is a solenoid with an iron core.	WS 3.1 Draw the magnetic field pattern for a solenoid carrying a current (showing the direction of the field). WS 1.4 Compare the advantages and disadvantages of permanent and electromagnets for particular uses.

4.6.3.5 The motor effect (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe how a magnet and a current-carrying conductor exert a force on one another and show that Fleming's left-hand rule represents the relative orientations of the force, the conductor and the magnetic field.	When a conductor carrying a current is placed in a magnetic field the magnet producing the field and the conductor exert a force on each other. This is called the motor effect. The direction of the force on the conductor is reversed if either the direction of the current or the direction of the magnetic field is reversed.	WS 1.2 Use Fleming's left-hand rule to predict the direction of the force on a conductor.

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Apply the equation that links the force on a conductor to the magnetic flux density, the current and the length of conductor to calculate the forces involved.	The size of the force on the conductor depends on: - the magnetic flux density - the current in the conductor - the length of conductor in the magnetic field. For a conductor at right angles to a magnetic field and carrying a current: $force = magnetic\ flux\ density \times current \times length$ $[F = B I l]$ force, F, in newtons, N magnetic flux density, B, in tesla, T current, I, in amperes, A (amp is acceptable for ampere) length, l, in metres, m	MS 3c Apply this equation, which is given on the Physics equations sheet. WS 3.3 Carry out and represent mathematical and statistical analysis.

4.6.3.6 Electric motors (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how this force is used to cause rotation in electric motors.	A simple electric motor consists of a rectangular coil of wire that is free to turn in the magnetic field of a permanent magnet. A commutator reverses the direction of the current every half turn, to allow the rotation to continue.	WS 1.2 Apply Fleming's left-hand rule to a simple electric motor. WS 1.4 Explain everyday and technological applications of science.